The component of velocity along vertical direction (y-axis) is $v_y = u_y + a_y t$

Since, $u_y = 0$, $a_y = g$, we get

$$v_y = gt \qquad \rightarrow (2.27)$$

Hence the velocity of the particle at any instant is

$$\vec{v} = u\hat{i} + gt\hat{j}$$

The speed of the particle at any instant t is given by

$$\therefore \quad v = \sqrt{v_x^2 + v_y^2}$$
$$v = \sqrt{u^2 + g^2 t^2}$$

(4) Speed of the projectile when it hits the ground: When the projectile hits the ground after initially thrown horizontally from the top of tower of height h, the time of flight is

$$T = \sqrt{\frac{2h}{g}}$$

The horizontal component velocity of the projectile remains the same i.e $v_x = u$

The vertical component velocity of the projectile at time T is

$$v_y = gT = g\sqrt{\frac{2h}{g}} = \sqrt{2gh}$$

The speed of the particle when it reaches the ground is

$$v = \sqrt{u^2 + 2gh}$$

2.11.3 Projectile under an angular projection

This projectile motion takes place when the initial velocity is not horizontal, but at some angle with the vertical, as shown in Figure 2.42.

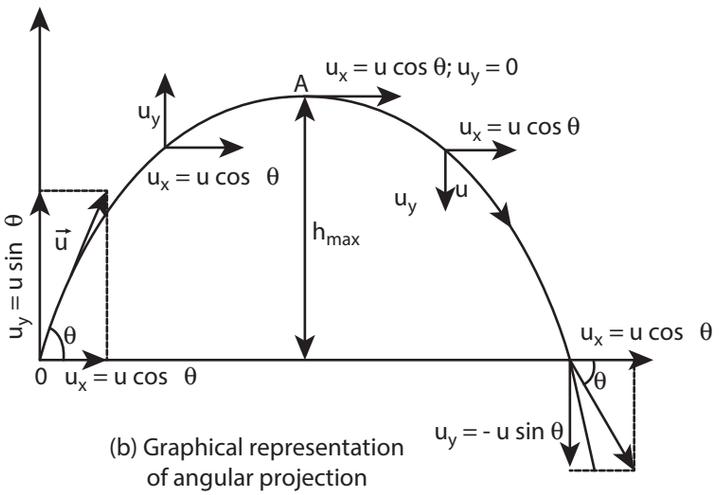
(Oblique projectile)

Examples:

- Water ejected out of a hose pipe held obliquely.
- Cannon fired in a battle ground.



(a) Water is ejected out of a pipe held obliquely



(b) Graphical representation of angular projection

Figure 2.42. Projectile motion

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